

Hackathon - 2025

**siRNA molecule based therapeutics
Biology**

Presented by:

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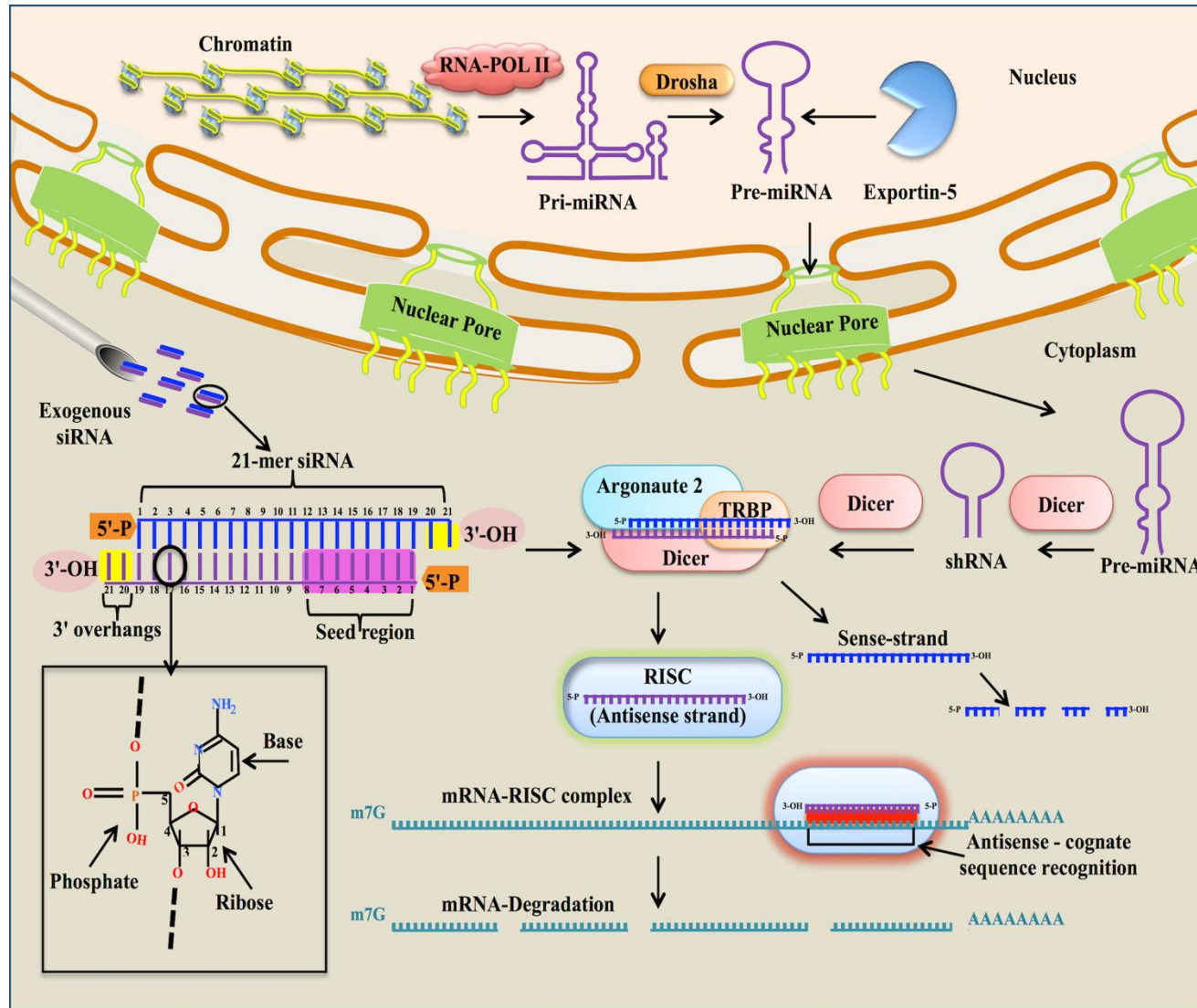
National Institutes of Health



Introduction

- ✓ Andrew Fire and co-workers described **RNA interference** (RNAi) phenomenon for first time in *C. elegans*
(A. Fire et al., Nature 1998)
- ✓ Double stranded RNA, ~21 nucleotide long potentially and specifically interferes to silence endogenous cognate mRNA
(A. Fire et al., Nature 1998; Filipowicz et al., Cell 2005)
- ✓ Small interfering RNA (siRNA) technology is used in functional genomics and development of potential therapeutics
(Woessmann et al., Reviews in clinical and expt. hematology 2003)
- ✓ More than 20 siRNA based therapeutics are in different clinical trials phases for leukemia, solid tumors, and viral infection
(Davis et al., Nature 2010;
Martínez et al., Experimental and Clinical Sciences journal 2015)
- ✓ RNA activation (similar to RNAi), directed to promoter region of target gene leads to its active transcription
(Janowski et al., Nature Chemical Biology 2007)

RNAi mechanism



Limitations:

- Delivery
- Efficacy
- Off-targets/Toxicity
- Nuclease digestion
- Cellular uptake
- Biological half-life
- Immunomodulation

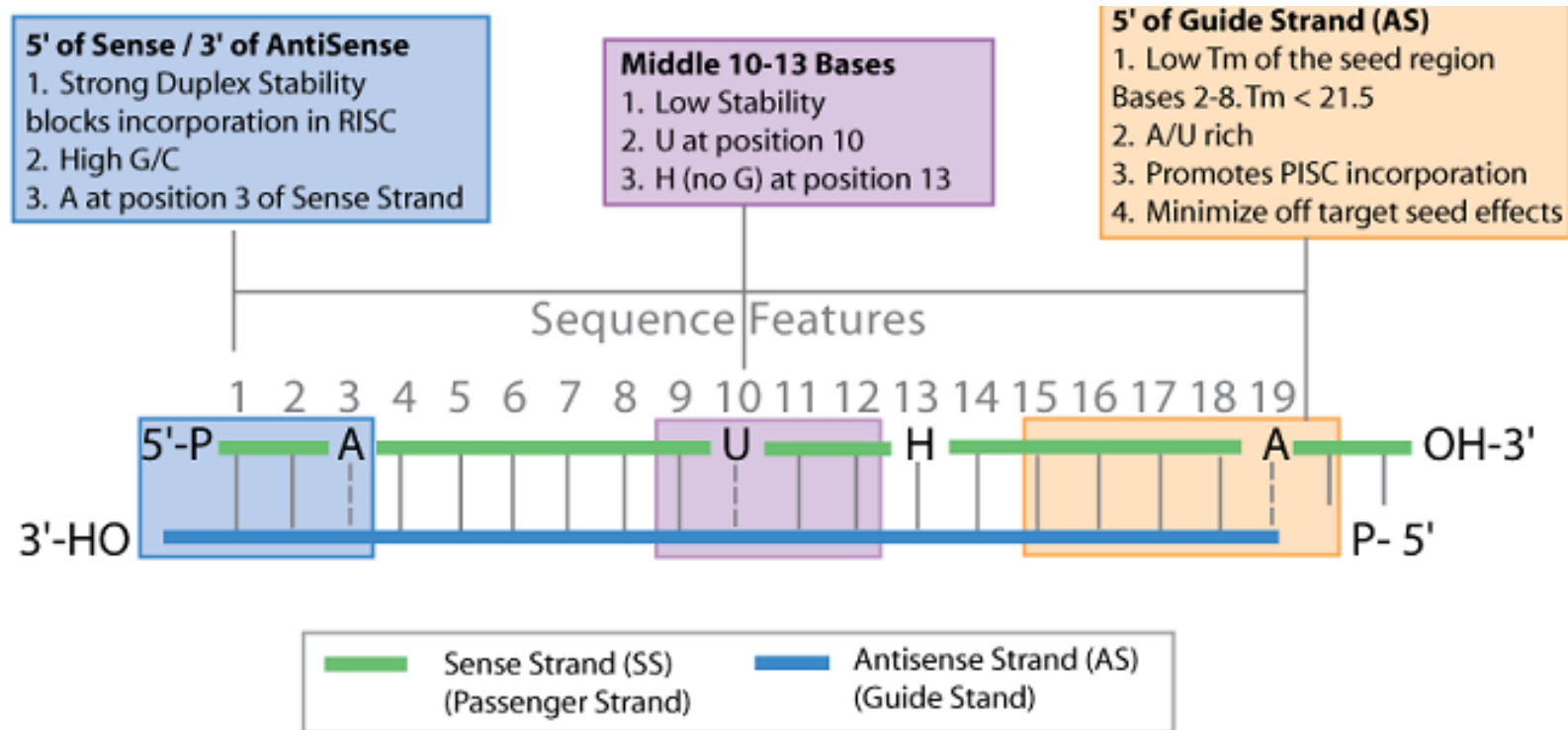
Joo et al., 2014 ; Sarett et al.2015
Journal of controlled release

Chemical modifications rescue these limitations ...

Property	Example Modification	Reference
Cellular uptake and distribution	Phosphorothioate	Detzer et al., 2008
Efficacy	Hexitol nucleic acid, Unlocked nucleic acid	Bramsen et al., 2009
Nuclease resistance	Locked Nucleic Acid; 2'-Fluoro modification	Glud et al., 2009
Immunomodulation	2'-O-Methyl ribose	Broering et al., 2014
Off-targets	Hypoxanthine; 2'-Thiouridine modification	Jackson et al., 2006; Takahashi et al., 2012
Delivery methods	Cholesterol	Yin et al., 2013

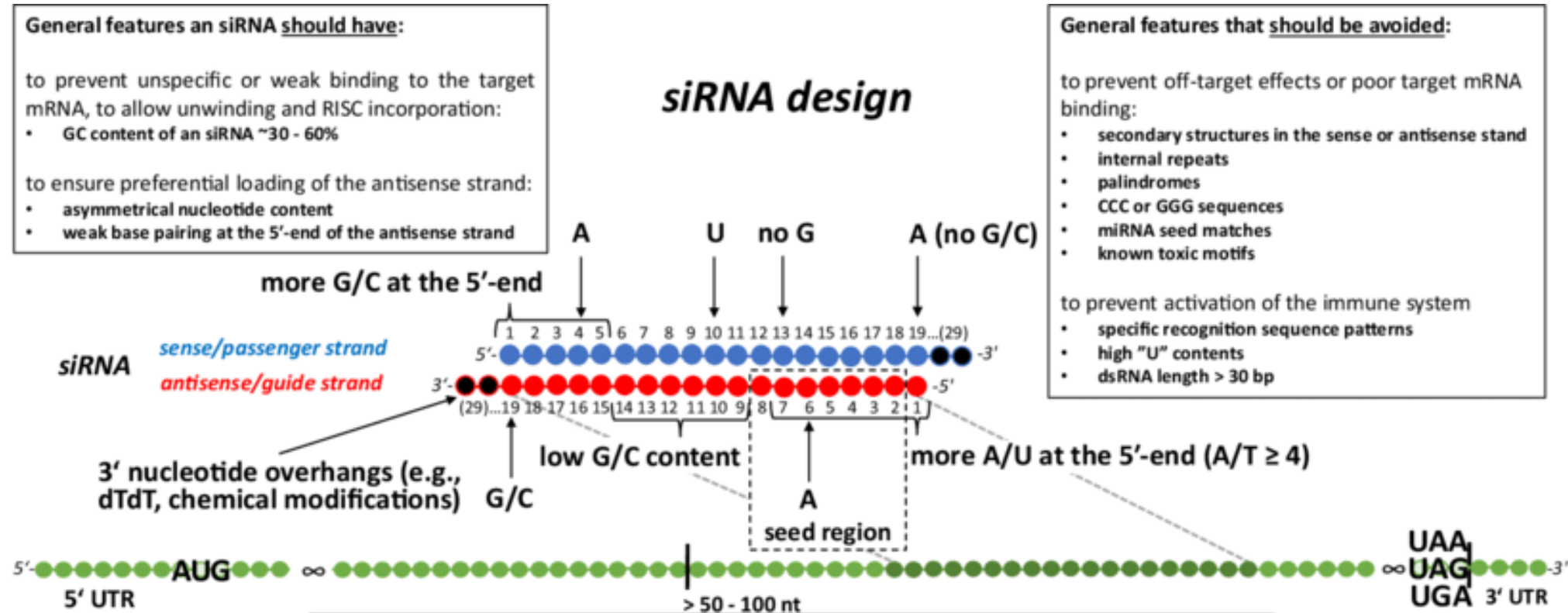


First generation – Feature based design

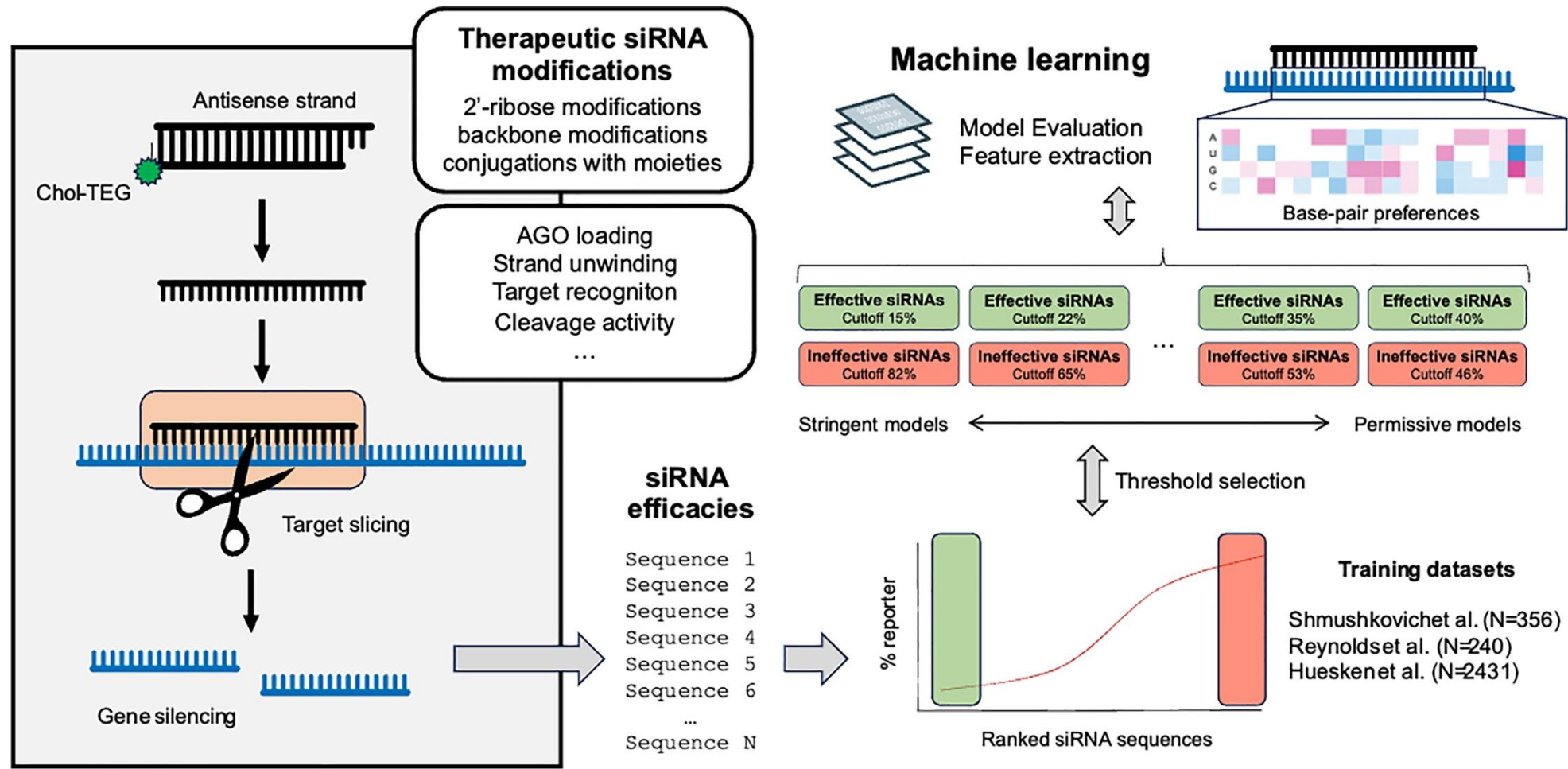


Individual character-based designing.

First generation – Feature based design (getting complicated)

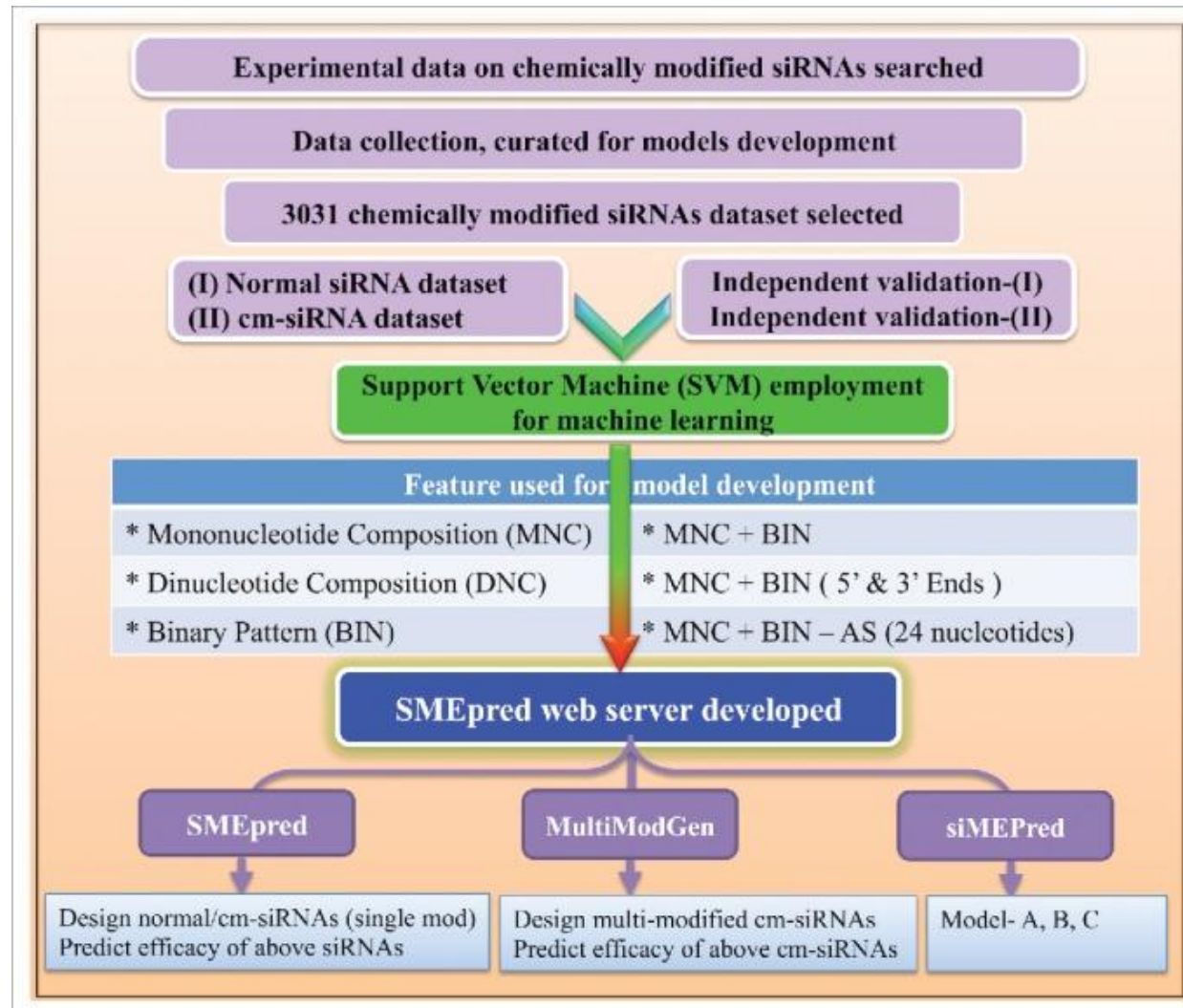


2nd generation siRNA design - Machine Learning



Source: Toward learning the rules that predict siRNA efficacy (Mol. Therapy Nucleic Acids)

3rd generation siRNA design - Machine Learning



Source: SMEpred workbench: A web server for predicting efficacy of chemically modified siRNAs

Step-1

- Data collection, curation – 3031 siRNAs (with 30 top modifications)

Step-2

- Feature selection of the dataset

Step-3

- Machine Learning Models

Step-4

- Model Development - 10-fold cross & independent validation

Step-5

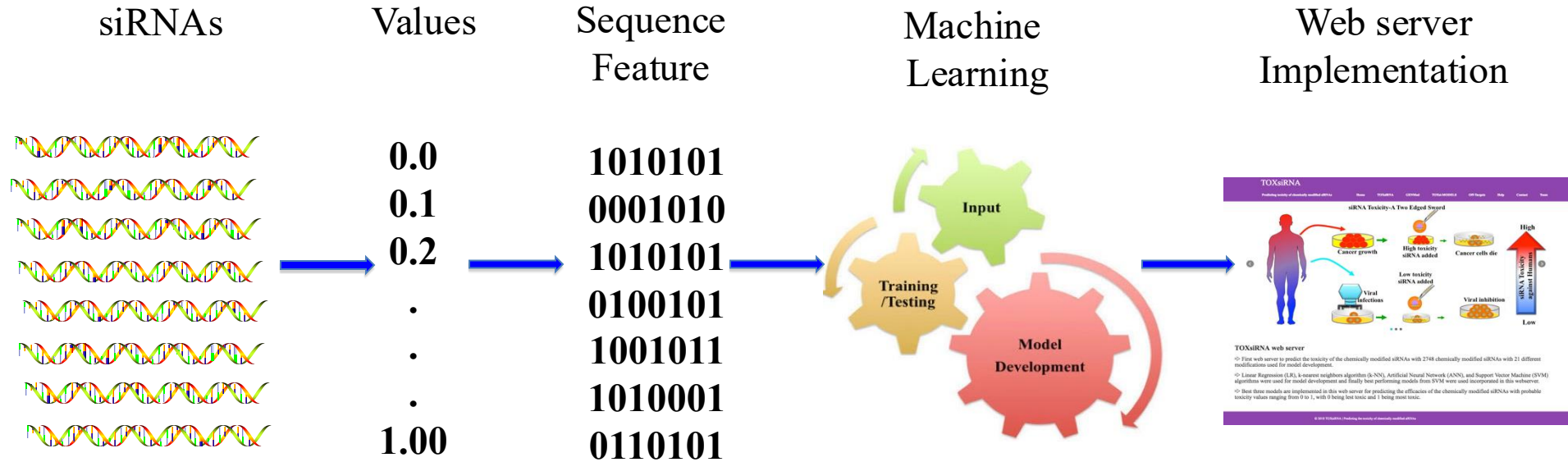
- Model(s) implementation

Chemical modifications

Chemical Modification	One letter Code	Binary code
Adenosine Nucleic Acid	A	000001
2'-N-Adamant-1-yl-Methylcarbonyl-2'-Amino-Locked Nucleic Acid	B	000110
Cytosine Nucleic Acid	C	000010
2'-Aminoethoxymethyl Nucleic Acid	D	000111
2'-Aminoethyl Nucleic Acid	E	001000
2'-Fluoro Nucleic Acid	F	001001
Guanosine Nucleic Acid	G	000011
4'-C-Hydroxymethyl Deoxyribonucleic acid	H	001010
Altritol Nucleic Acid	I	001011
2'-Aminopropyl Nucleic Acid	J	001100
2'-aminopropoxymethyl Nucleic Acid	K	001101
Locked Nucleic Acid	L	001110
2'-Deoxy-2'-N,4'-C-Ethylene-Locked Nucleic Acid	M	001111
Alfa-L-Locked Nucleic Acid	N	010000
2'-Cyanoethyl Nucleic Acid	O	010001
2',4'-Carbocyclic-Ethylene-bridged Nucleic Acid-Locked Nucleic Acid	P	010010
2',4'-Carbocyclic-Locked Nucleic Acid-Locked Nucleic Acid	Q	010011

Chemical Modification	One letter Code	Binary code
2'-Guanidinoethyl Nucleic Acid	R	010100
Hexitol Nucleic Acid	S	010101
Thymidine Nucleic Acid	T	000100
Uracil nucleic acid	U	000101
Oxetane-Locked Nucleic Acid	V	010111
2'-N-Pyren-1-yl Methyl-2'-Amino-Locked Nucleic Acid	W	011000
Unlocked Nucleic Acid	X	011001
2'-O-Methyl Nucleic Acid	Y	011010
2'-Deoxy Nucleic Acid	Z	011011
2'-Deoxy-2'-Fluoro Nucleic Acid	b	011100
2'-O-Benzyl Nucleic Acid	d	011101
2'-O-Guanidinopropyl Nucleic Acid	e	011110
Boranophosphate Nucleic Acid	g	011111
Dihydrouridine Nucleic Acid	h	100000
Hypoxanthine Nucleic Acid	m	100001
Peptide Nucleic Acids	n	100010
Phosphorothioate Nucleic Acid	q	100011
Tricyclodeoxyribo Nucleic Acid	r	100100

Methodology



- siRNAs
- Modifications
- ML/AI Models
 - I. Linear Regression (LR)
 - II. K-Nearest Neighbors (KNN)
 - III. Artificial Neural Networks (ANN)
 - IV. Support Vector Machine (SVM)
 - V. Random forest
 - VI. 1D CNNs
 - VII. Gradient Boost

Comparison of Machine learning models in terms of Pearsons Correlation Coefficient (PCC)

	Feature	Vector size	LR	KNN	ANN	SVM
1	Mononucleotide (A) (I)	52	0.84	0.83	0.83	0.91
2	Binary(B)	48	0.59	0.8	0.79	0.34
3	Dinucleotide	1352	0.86	0.85	0.15	0.88
4	A+B 24-ntd Both strands	100	0.78	0.83	0.81	0.33
5	A+B 16-ntd Both strands	84	0.89	0.82	0.8	0.61
6	A+B 13ntd Both strands	78	0.79	0.82	0.81	0.62
7	A+B 8-ntd Both strands	68	0.82	0.82	0.82	0.71
8	A+B 24-ntd Antisense only (II)	76	0.86	0.83	0.83	0.86
9	A+B 16-ntd Antisense only	68	0.8	0.83	0.83	0.88
10	A+B 13ntd Antisense only	65	0.8	0.83	0.79	0.88
11	A+B 8-ntd Antisense only	60	0.82	0.83	0.82	0.89
12	A+B 24-ntd Sense only (III)	76	0.86	0.82	0.78	0.88
13	A+B 16-ntd Sense only	68	0.88	0.81	0.8	0.89
14	A+B 13ntd Sense only	65	0.83	0.82	0.8	0.89
15	A+B 8-ntd Sense only	60	0.83	0.82	0.8	0.9

LR= Linear regression;
ANN =Artificial neural networks;

KNN = K-nearest neighbors;
SVM= support vector machine

Color Scale 
High Low

Example Data...



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